

Marius Arcay

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EDUCATION

San Jose State University

Bachelor of Science in Computer Science

San Jose, CA

Spring 2026 – Spring 2028 (Expected)

West Valley College

Associate of Science in Physics

Associate of Science in Mathematics

Saratoga, CA

Winter 2026 – Spring 2028 (Expected)

Fall 2022 – Winter 2026

EXPERIENCE

Contract 3D Artist

Oct 2025 – Jan 2026

Valve Corporation

Remote

- Authored [Rooftop](#), an official Counter-Strike 2 multiplayer map using Valve's proprietary technology, balancing visual fidelity, gameplay, and performance
- Built, profiled and optimized scene geometry, materials, lighting, and assets to meet performance standards expected of an official Counter-Strike 2 map
- Shipped to a global audience of millions of players

Retail Tech Consultant

Jun 2021 – Present

Target

Morgan Hill, CA

- Provided personalized expertise to help guests select, understand, and purchase electronics

PROJECTS

[CUDA Fractal Renderer](#) [\[Technical Write-up\]](#) | C++, CUDA, Multi-threading, Git, Visual Studio

- Built a CUDA fractal renderer achieving a 100× GPU speedup over single-threaded CPU benchmarks across Mandelbrot and Julia sets
- Developed a benchmarking system comparing single-threaded CPU, multi-threaded CPU, and CUDA rendering speeds, alongside dynamic iteration scaling, click-and-drag interactions, and a video/animation rendering pipeline
- Automated Visual Studio project configuration to detect unsupported CUDA environments and remove CUDA-specific dependencies during compilation

[OpenGL Path Tracing Engine](#) [\[Technical Write-up\]](#) | C++, OpenGL, GLSL, Git, Visual Studio

- Built a GPU-accelerated path tracer using OpenGL/GLSL with Monte Carlo sampling and Cook-Torrance GGX PBR, supporting albedo, normal, roughness, and metallic maps
- Implemented a depth prepass to reduce overdraw alongside scene and model importing, texture pooling, bounding-volume intersection testing, animation system, and image/video rendering
- Developed interactive rendering tools including a translation gizmo, real-time material editing, debug visualization modes, and scene manipulation controls

[3D Graphics Library](#) | C++, Multi-threading, Visual Studio

- Built a multi-threaded CPU 3D renderer in C++
- Implemented a model conversion pipeline, simple animation system, and flexible scene configuration

[Environment Monitoring Device](#) | C, MySQL, I2C

- Developed a Raspberry Pi-based device in C to collect temperature, humidity, and luminance measurements through I2C sensors
- Implemented MySQL-based logging and configurable data visualization for long-term environmental monitoring

TECHNICAL SKILLS

Languages: C++, C, GLSL, Python, Java, Assembly (ARM)

Graphics & GPU: OpenGL, CUDA, PBR, Monte Carlo Rendering

Systems: Multithreading, Linux, Windows, I2C

Libraries & Tools: OpenCV, raylib, Git, Visual Studio, VS Code